



CHAYMA BETTAHER

Final-Year Software Engineering Student

Tunis, Tunisia | **Phone:** +216 55 684 933 | **Email:** bettaherchayma77@gmail.com

GitHub: <https://github.com/Bchayma> **LinkedIn:** [chayma bettaher](#)

PROFILE

Final-year Software Engineering student (5th year) at the Faculty of Sciences of Tunis, with hands-on experience in full-stack development, cloud computing, DevOps, and applied machine learning. Built a strong foundation in mathematics, algorithmics, and physics through an integrated preparatory cycle before specializing in software engineering. Particularly interested in artificial intelligence, machine learning, cloud computing, and software engineering research. Comfortable working independently and in multidisciplinary teams, with strong communication skills in French, English, and Arabic.

EDUCATION

Software Engineering (Cycle Ingénieur) | Faculty of Sciences of Tunis 2022 - Present

5th and final year of Software Engineering studies (expected graduation: 2027)

- Integrated Preparatory Cycle (2022-2024): intensive foundation in Advanced Mathematics (analysis, algebra, probability), Algorithmics, Physics, and core Computer Science
- Software Engineering Cycle (2024-Present): specialization in application development, data engineering, DevOps, and AI/ML

Mathematics Baccalaureate - Highest Honors | Pioneer High School of Ariana 2018 - 2022

PROFESSIONAL EXPERIENCE

IT Consultant | Optima Junior Enterprise 2025 - 2026

- Development of websites using WordPress and WooCommerce
- SEO and web performance optimization
- Prospecting, negotiation, and client relationship management

Full-Stack Developer Intern | NADA Groups July 2026

- Developed a full-stack healthcare management application using Next.js, NestJS, and MongoDB/Mongoose
- Designed RESTful APIs and implemented JWT authentication and role-based access control for patients, doctors, administrators, and pharmacies
- Developed dashboards and integrated frontend/backend services for managing patients, prescriptions, appointments, medications, pharmacies, and stock

Technologies: Next.js, NestJS, MongoDB, Mongoose, JWT, Docker, TypeScript

Cybersecurity & Machine Learning Internship | VOCALCOM August 2025

- Development of a network anomaly detection system
- Network traffic analysis using Wireshark
- Machine learning model implementation
- Creation of a Streamlit dashboard for anomaly visualization

Technologies: Python, Scikit-learn, Pandas, Streamlit, Wireshark, ML

DevOps & Kubernetes Internship | LOOYAS June - July 2025

- Implementation of CI/CD pipelines using Jenkins and GitHub
- Containerization with Docker and orchestration with Kubernetes
- Configuration of Istio and Nexus Repository
- Deployment automation and Infrastructure as Code

Technologies: Jenkins, Docker, Kubernetes, Istio, Nexus, CI/CD

TECHNICAL PROJECTS

Intelligent Exam Management System

- Web application for automated university exam management
- Full-stack development using Angular, Spring Boot, and PostgreSQL
- Implementation of Dependency Injection, transaction management, and layered architecture with Spring

Technologies: Angular, Spring Boot, Spring Data JPA, Hibernate, PostgreSQL, Maven, REST API

Deep Learning Project - Anomaly Detection with Convolutional Autoencoder

- Developed an anomaly detection system based on a Convolutional Autoencoder (CAE) applied to the Fashion-MNIST dataset
- Trained the model on normal data and detected anomalies through reconstruction error
- Implemented and optimized Deep Learning models using PyTorch and Optuna

Technologies: Python, PyTorch, Scikit-learn, Optuna, NumPy, Matplotlib, Deep Learning, Autoencoders

Blockchain Project - SkillNFT

- Development of a blockchain platform for skill certification via NFTs
- Implementation of Solidity smart contracts using the ERC-721 standard
- Secure management of digital certificates with MetaMask and EVM blockchain

Technologies: Solidity, React, MetaMask, Ethereum, ERC-721, Remix, Foundry

Mobile Movie Management Application

- Development of a Flutter mobile application for movie management with user authentication
- Integration of the MovieDB API and Firebase data storage
- User management and movie administration

Technologies: Flutter, Dart, Firebase, Firestore, REST API, MovieDB API

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, Java, PHP, C/C++, SQL, HTML/CSS, Bash

Web & Mobile Development: Angular, React, Next.js, NestJS, Flutter, Spring Boot, Node.js, Express, Symfony, REST APIs, WordPress

Databases: PostgreSQL, MySQL, SQL Server, Oracle, MongoDB, Redis, Hibernate, JPA

DevOps & Cloud: Docker, Kubernetes, Jenkins, GitHub Actions, Istio, Nexus, AWS

Data Engineering & BI: SSIS, ETL, SQL Server, Data Warehouse, Power BI, Data Visualization

Machine Learning & Data Science: Scikit-learn, TensorFlow, Keras, PyTorch, NLP, Deep Learning, Pandas, NumPy

Cybersecurity: Wireshark, anomaly detection, web security

Tools & Methodologies: Git, GitHub, Postman, GraphQL, Scrum, Agile, Maven

Mathematics & Algorithmics: Advanced Mathematics, Advanced Algorithmics, Data Structures, Graph Theory, Complexity Optimization, Dynamic Programming

CERTIFICATIONS

AWS Certified Cloud Practitioner | Amazon Web Services

- Fundamental concepts of Cloud Computing and AWS services
- Mastery of AWS services: EC2, S3, RDS, Lambda, VPC, IAM
- Security, scalability, and cloud architecture

Fundamentals of Deep Learning | NVIDIA

- Fundamental concepts of Deep Learning and neural networks
- Model training and optimization with GPU
- Introduction to Deep Learning and AI frameworks

INTERESTS & ASSOCIATIVE LIFE

Logistics Manager | Tunisian Association of Artificial Intelligence (ATIA)

- Team, partner, and sponsor management

Competitions & Hackathons

- Participation in workshops and technical events
- Competitive programming and algorithmic problem solving

SOFT SKILLS

Communication • Teamwork • Problem Solving • Leadership • Time Management

LANGUAGES

Arabic: Native | French: Fluent (C1) | English: Fluent (B2-C1) | Spanish: Intermediate (B1)